MIT 18.06SC - [redacted] 1.6

Problem 5.1)

a) $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Since the above matrix moves the first row to the second rows position, the second to the third's and the third to the first, cubing P will give the identity matrix as the above cycle will be repeated thrice

b) $\hat{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$\hat{P} \neq I$ because the first row and column will always stay the same regardless of the power $\hat{P}$ is raised to, and the remaining matrix is then identical to ~~P from p~~ matrix P from part (a), and so $\hat{P}^3 = I$, and so $\hat{P}^4 \neq I$

Problem 5.2)

a) If A is symmetric, we can independently choose all the elements along the main diagonal (i.e 4 elements) and all the elements either above or below the diagonal (i.e 6 elements)

The remaining entries can of the diagonal not be chosen independently as they ~~are~~ must match the entries on the opposite side which was filled first, and so we get a total of $6+4=10$ total entries that can be independently chosen.

b) If A is skew symmetric, the ~~diagonal~~ entries of the main diagonal must be 0 and so can not be independently chosen; and the entries on each side of the main diagonal must be the negative of the coresponding entries on the other side i.e ~~one only of~~ only the entries either above or below can be independently chosen. i.e ~~or only~~ 6 entries in total can be independently chosen.

Problem 5.3)

a) Let A and B be any ~~s~~ two symmetric matrices. Then $(A+B)^T = A^T + B^T = A+B$, and so $A+B$ is also symmetrical (i.e the set is closed under addition). Let x be a scalar, then $(xA)^T = xA^T = xA$ and so the set is also closed under multiplication, ~~and~~ so the statement is TRUE

b) Let A and B be any two skew symmetric matrices, then $(A+B)^T = A^T + B^T = -A-B = -(A+B)$ and so the set is closed under addition. Let x be

any scalar, the $(xA)^T = x \times -A = -xA = -(xA)$ and so the set is closed under multiplication. Hence the statement is ~~True~~ TRUE

c) Let $A = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 0 & 3 \end{bmatrix}$

Then $A+B = \begin{bmatrix} 2 & 2 \\ 2 & 8 \end{bmatrix}$

Since A and B are not symmetric but A+B is, the set is not closed under addition and so the statement is ~~false~~ FALSE